

PAF INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY · DEPARTMENT OF
COMPUTER SCIENCE & ARTIFICIAL INTELLIGENCE

COMP-341 Assignment 02 Report

Hebbian / Neo-Hebbian Learning and STDP

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Course

Artificial Neural Network (COMP-341)

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1. Introduction

This report presents the implementation and analysis of the assignment tasks related to Hebbian learning, Oja's rule, principal component extraction, and Spike-Timing Dependent Plasticity (STDP). The work is organized exactly by assignment tasks and evaluated with emphasis on:

- conceptual correctness,
- code-to-theory consistency,
- plot quality and interpretation,
- mathematically consistent notation.

The report includes only essential code logic and focuses primarily on method, outputs, and interpretation.

2. Q1: Oja's Rule and Unsupervised PCA (20 Marks)

2.1 Task 2.1 - Basic Hebbian Rule (Unstable) [5 Marks]

2.1.1 Task Objective

Implement the classical Hebbian update on correlated 2D data and demonstrate instability of the weight dynamics.

2.1.2 Core Method

Basic Hebbian update used:

$$\Delta w = \eta y x, \quad y = w^T x$$

Minimal implementation idea:

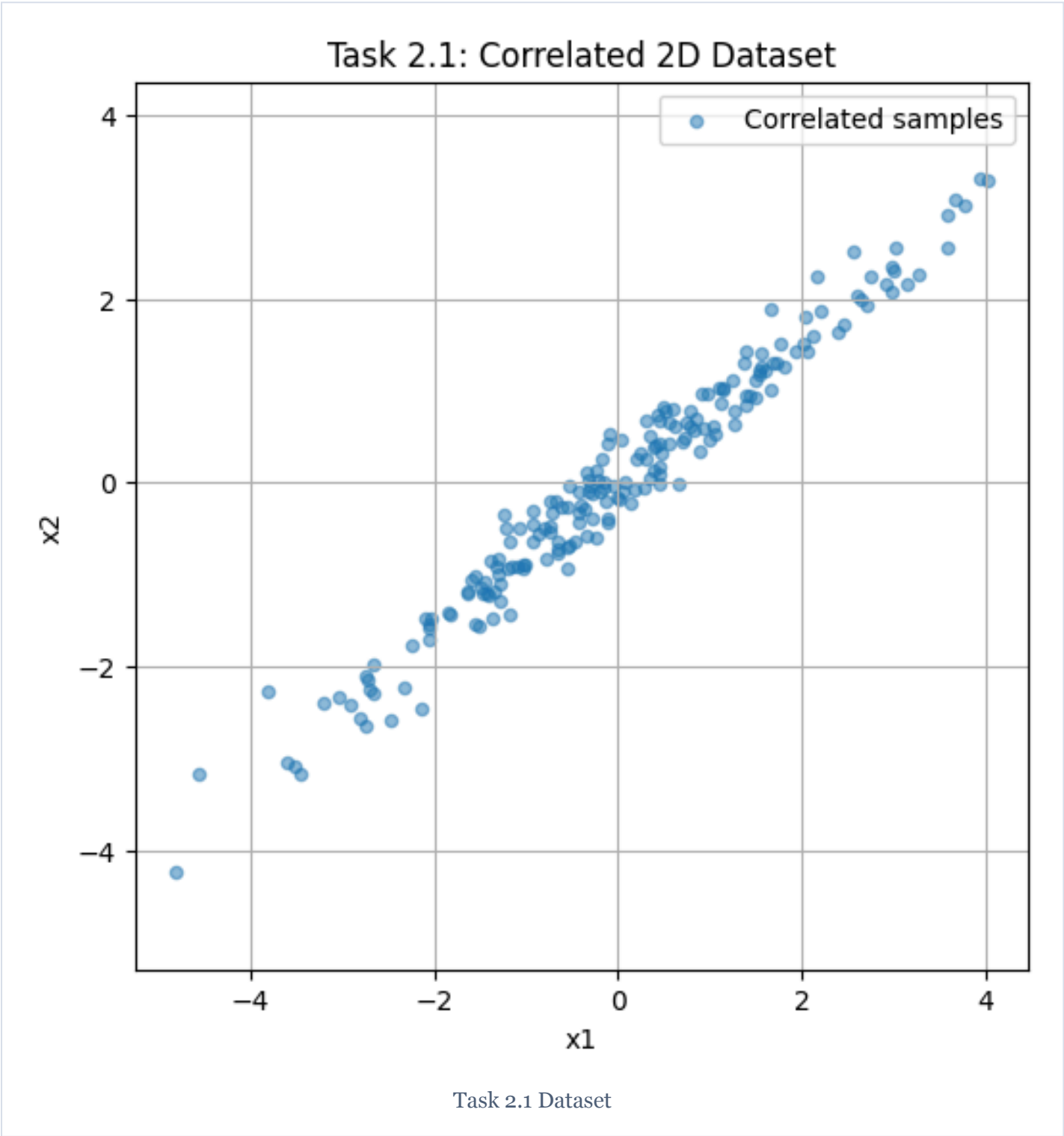
```
for x in X:
    y = np.dot(w, x)
    w += eta * y * x
```

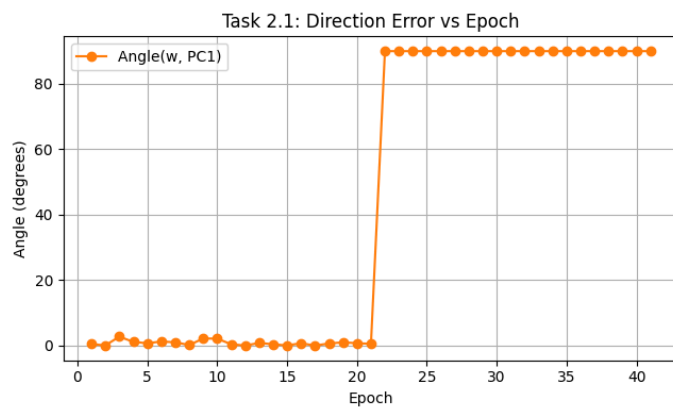
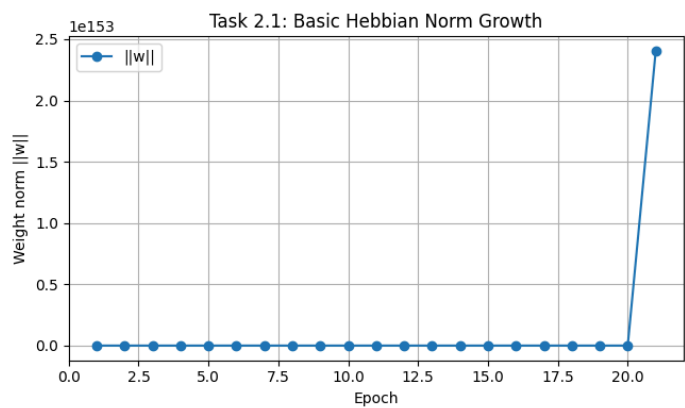
2.1.3 Data and Observed Outputs

```
Dataset shape: (200, 2)
Empirical covariance matrix:
[[2.95632304 2.3439734 ]
 [2.3439734  1.92483448]]

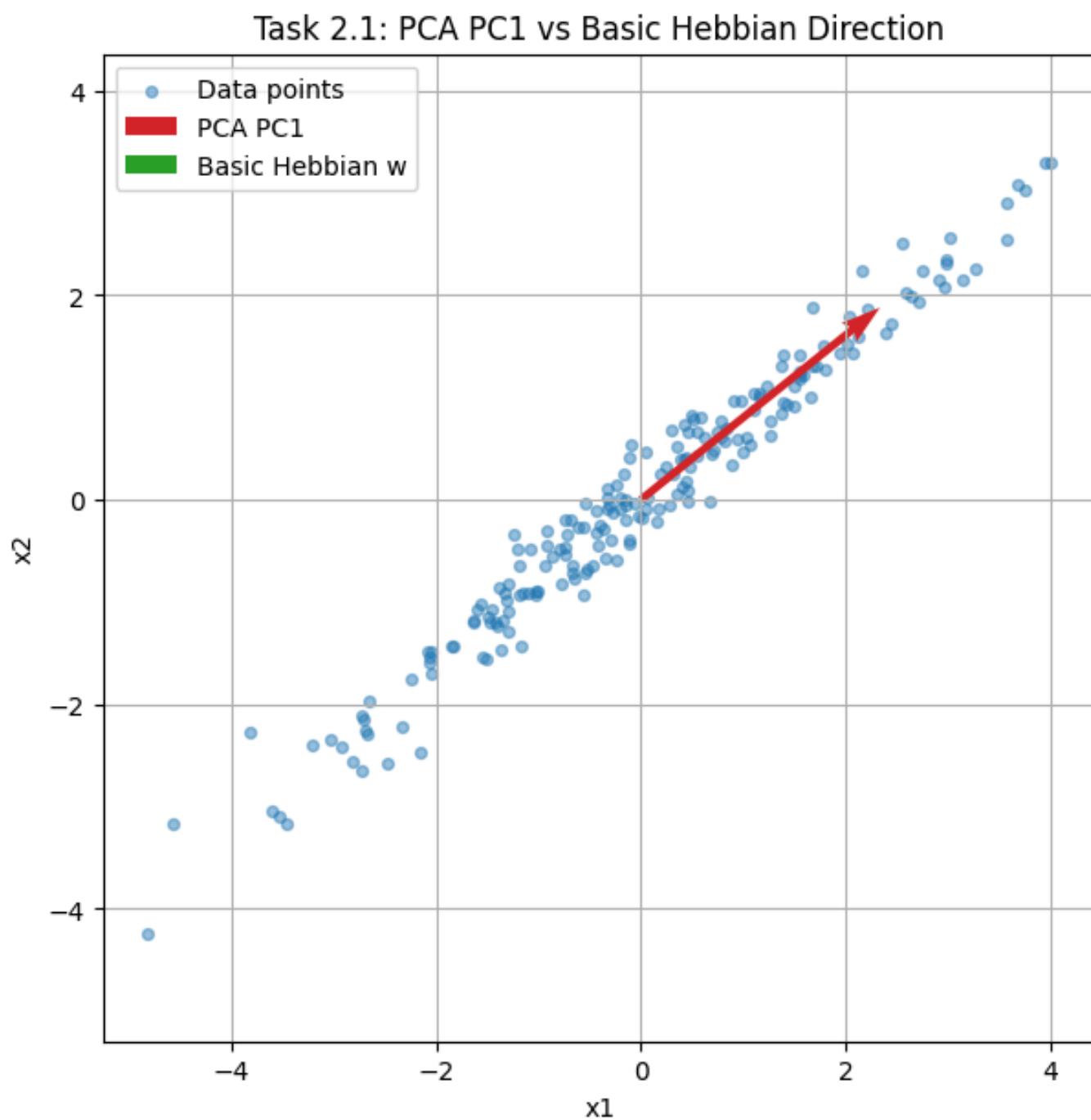
Basic Hebbian final alignment: angle=nan deg, |cosine|=nan
Final ||w||: nan
```

2.1.4 Visual Results





Task 2.1 Instability Curves



Task 2.1 PCA vs Basic Hebbian Direction

2.1.5 Interpretation

The run diverged numerically (`nan`), which is consistent with the expected instability of unconstrained Hebbian growth on correlated inputs. Since there is no normalization/competition term, weight norm can blow up, and direction comparison with true PC becomes invalid. This directly supports the assignment requirement to show unstable behavior and lack of reliable principal-component convergence.

2.2 Task 2.2 - Oja's Rule (Stable PCA) [10 Marks]

2.2.1 Task Objective

Implement Oja's rule and verify convergence toward the first principal component obtained from sklearn PCA.

2.2.2 Core Method

Oja update used:

$$\Delta w = \eta y(x - yw)$$

Equivalent decomposition:

$$\Delta w = \eta(yx) - \eta(y^2w)$$

The second term acts as normalization/stabilization.

Minimal implementation idea:

```
for x in X:
    y = np.dot(w, x)
    w += eta * y * (x - y * w)
```

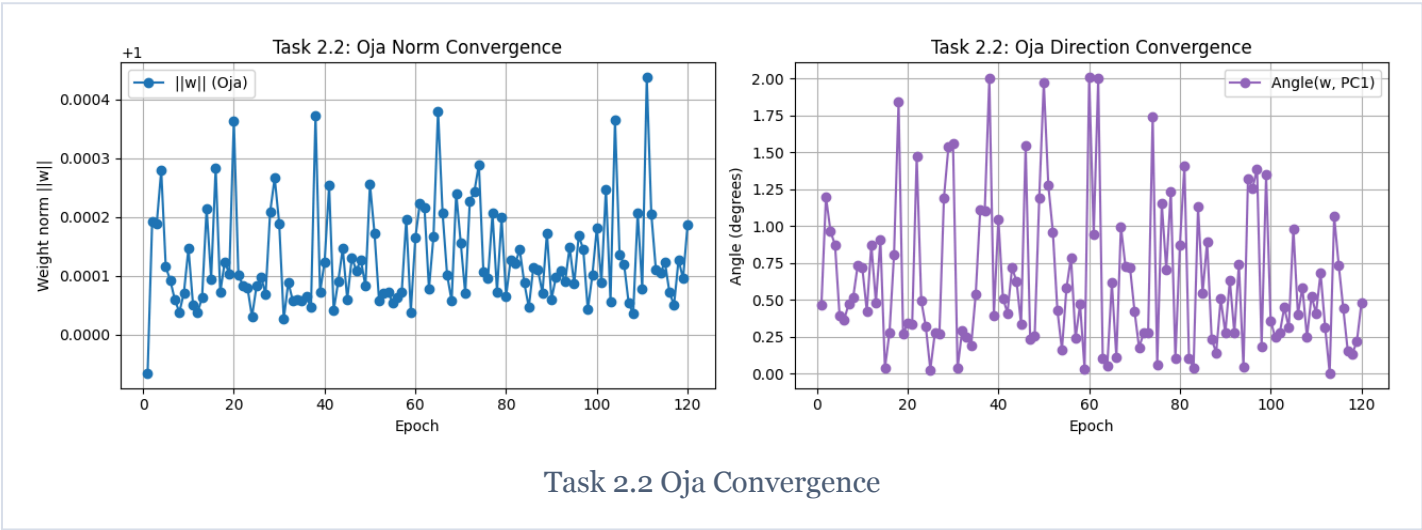
2.2.3 Observed Outputs

Oja final alignment: angle=0.4779 deg, |cosine|=0.999965
Final ||w|| before final unit-normalization (last epoch): 1.0001874864718119

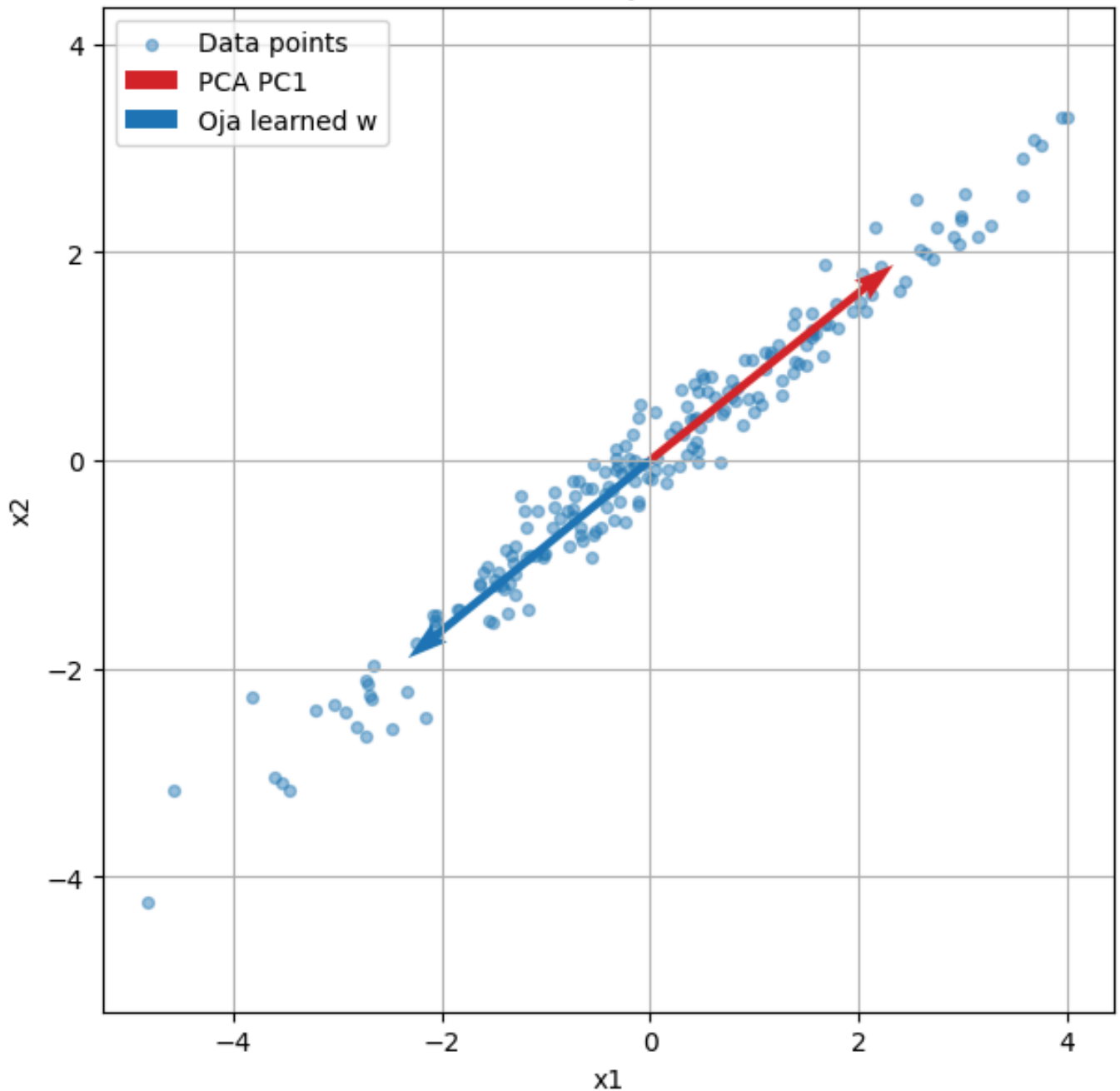
Comparison Summary

Basic Hebbian: final ||w|| = nan, final angle to PC1 = nan deg
Oja's Rule : final ||w|| ~ 1.000000 (unit), final angle to PC1 = 0.4779 deg

2.2.4 Visual Results



Task 2.2: PCA PC1 vs Oja Learned Direction



Task 2.2 PCA vs Oja Direction

2.2.5 Interpretation

The very small angle to PCA PC1 (0.4779°) and unit-scale norm confirm stable convergence to the dominant principal direction. This result validates Oja's rule as a stable unsupervised PCA-like learner.

2.2.6 Why Oja is Neo-Hebbian

Oja's rule preserves the Hebbian correlation term but augments it with a local stabilizing term proportional to $y^2 w$. Therefore, it is a modified Hebbian rule (Neo-Hebbian), not purely classical Hebbian.

2.3 Task 2.3 - Receptive Fields from Natural Image Patches [5 Marks]

2.3.1 Task Objective

Learn first 10 principal directions from 10,000 random 8×8 patches and analyze visual structure of learned weights.

2.3.2 Core Method

- Source patches from `olivetti_faces`.
- Flatten each 8×8 patch to a 64D vector.
- Mean-center/standardize.
- Learn components sequentially with Oja + deflation.

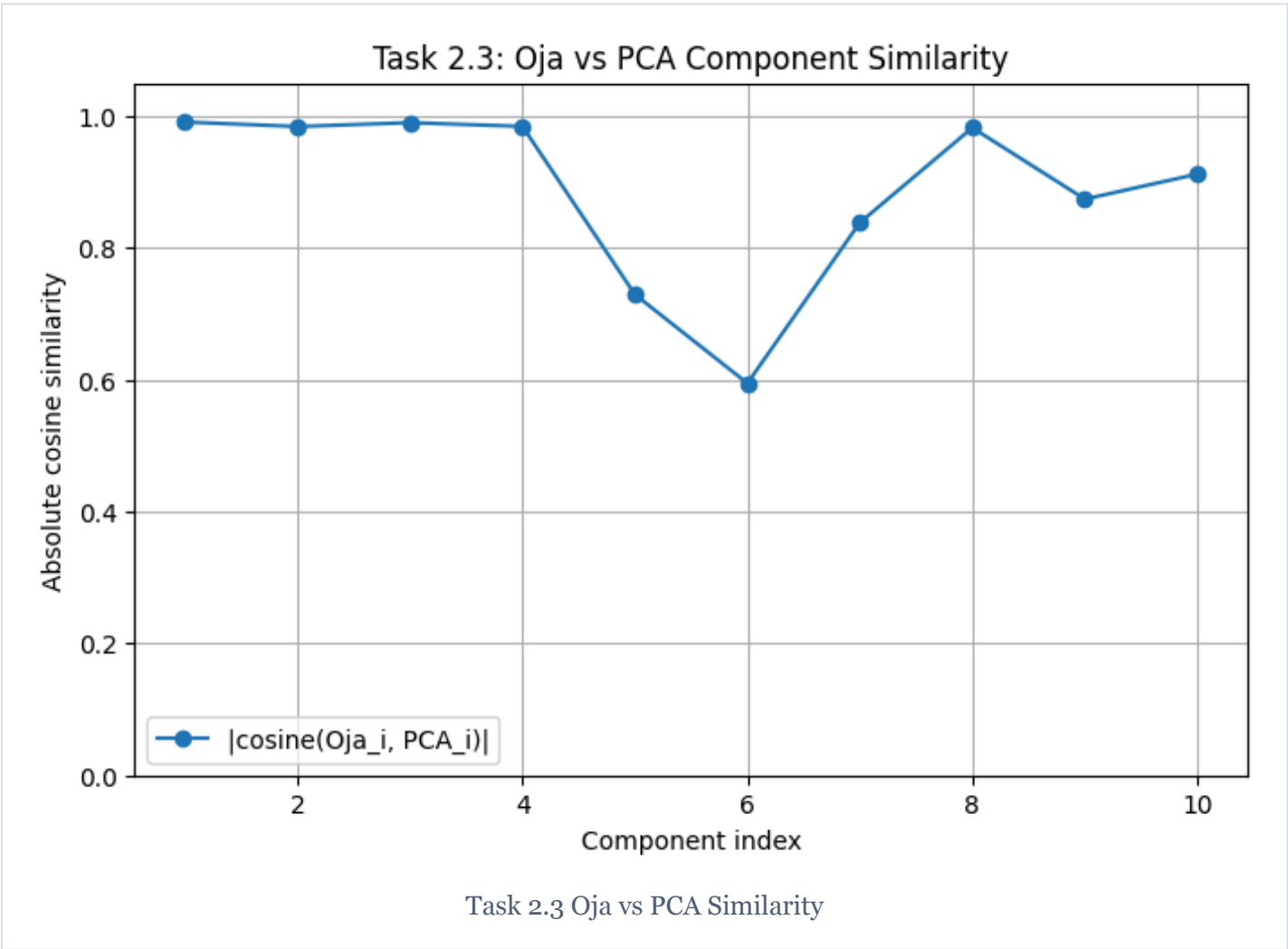
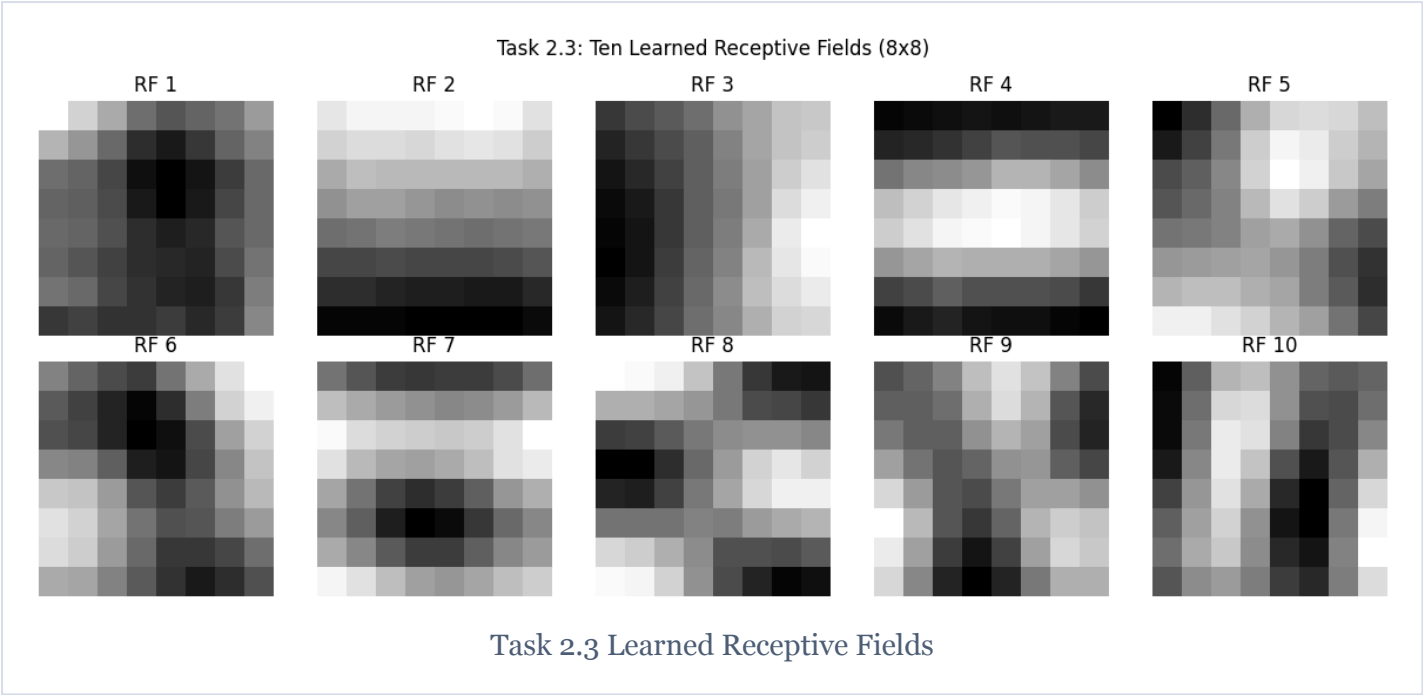
Essential multi-component logic:

```
for i in range(n_components):  
    w = ojas_rule(residual_X)  
    components.append(w)  
    residual_X = residual_X - np.outer(residual_X @ w, w)
```

2.3.3 Observed Outputs

```
Patch source: olivetti_faces  
Patch matrix shape: (10000, 64)  
Learned components shape: (10, 64)  
  
Mean similarity over 10 components: 0.888234999486834
```

2.3.4 Visual Results



2.3.5 Interpretation

The learned components show oriented and contrast-sensitive structures typical of natural-image statistics. Quantitatively, the high mean cosine similarity (0.8882) between Oja-learned and PCA

components supports the expected Hebbian->Oja->PCA linkage. Qualitatively, several filters are edge-like and resemble Gabor-like receptive-field behavior often discussed for V1.

3. Q3: Differential Hebbian Learning and Neo-Hebbian Extensions (20 Marks)

3.1 Part A - Theory [8 Marks]

3.1.1 Q3(a) Differential Hebbian Learning

Standard Hebbian rule:

$$\frac{dw}{dt} = \eta x(t)y(t)$$

Differential Hebbian rule:

$$\frac{dw}{dt} = \eta \frac{dx}{dt} \frac{dy}{dt}$$

Difference: standard Hebbian uses instantaneous co-activation; differential Hebbian emphasizes temporal co-variation (changes), so it is better aligned with predictive timing relationships.

3.1.2 Q3(b) STDP and Temporal Window

Define timing difference $\Delta t = t_{\text{post}} - t_{\text{pre}}$. A standard pair-based model is:

$$\Delta w = \begin{cases} A * +e^{-|\Delta t|/\tau^{*+}}, & \Delta t > 0 \quad (\text{LTP}) \\ -A * -e^{-|\Delta t|/\tau^{*-}}, & \Delta t < 0 \quad (\text{LTD}) \end{cases}$$

So if pre-synaptic firing precedes post-synaptic firing, potentiation occurs; for reverse order, depression occurs.

3.1.3 Q3(c) Mexican-Hat Correlation and Orientation Selectivity

A Mexican-hat correlation profile (local excitation, surround inhibition) encourages structured positive and negative receptive subregions under Hebbian-like adaptation. This supports emergence of orientation-selective filters in Linsker-style self-organization models.

3.2 Part B - Python STDP Simulation [12 Marks]

3.2.1 Task Objective

Simulate STDP using required parameters and compare random, causal, and anti-causal timing conditions.

3.2.2 Parameters Used

- $A_+ = 0.01$
- $A_- = 0.012$
- $\tau_+ = 20\text{ ms}$
- $\tau_- = 20\text{ ms}$
- $T = 1000\text{ ms}$
- epochs = 50

3.2.3 Essential Window Logic

```
if dt > 0:
    dw = A_plus * exp(-abs(dt)/tau_plus)
elif dt < 0:
    dw = -A_minus * exp(-abs(dt)/tau_minus)
```

3.2.4 Observed Outputs

Random case final weight: 0.9994316929072757

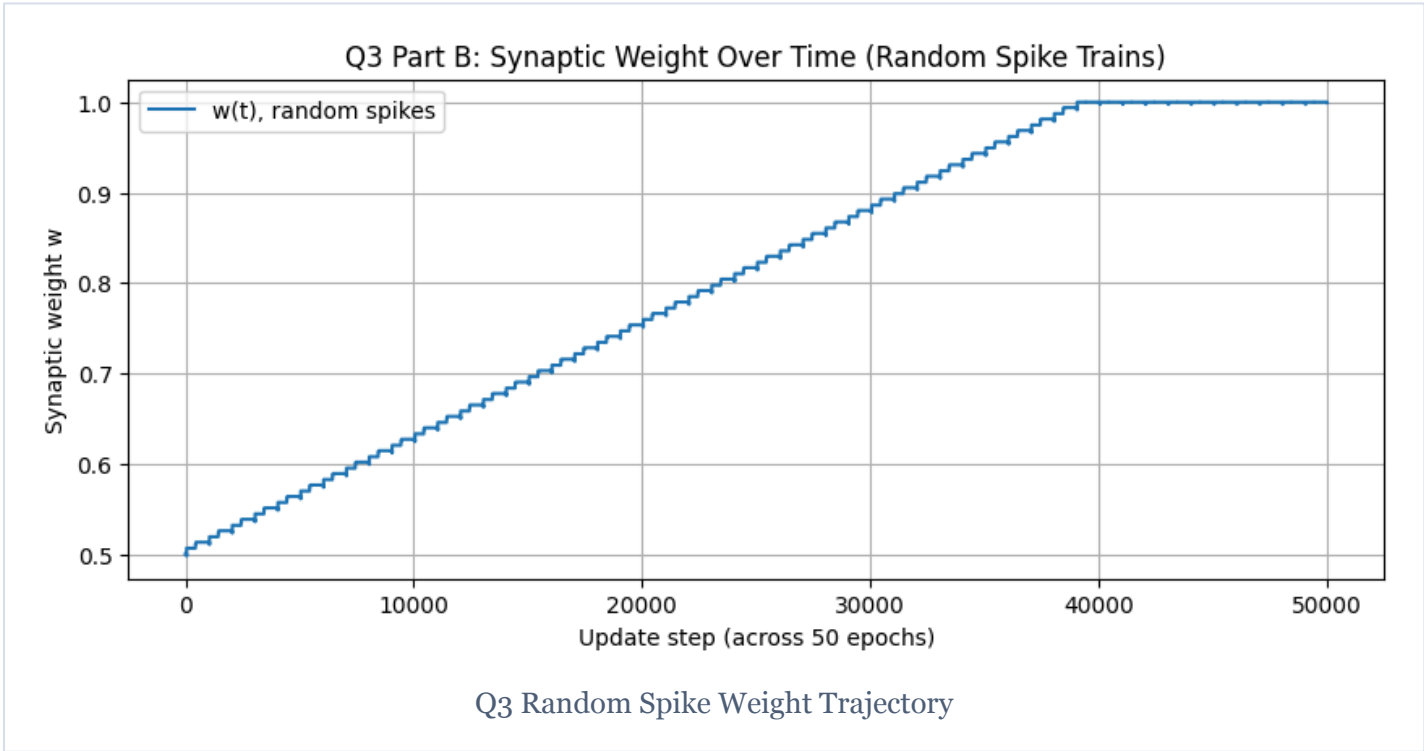
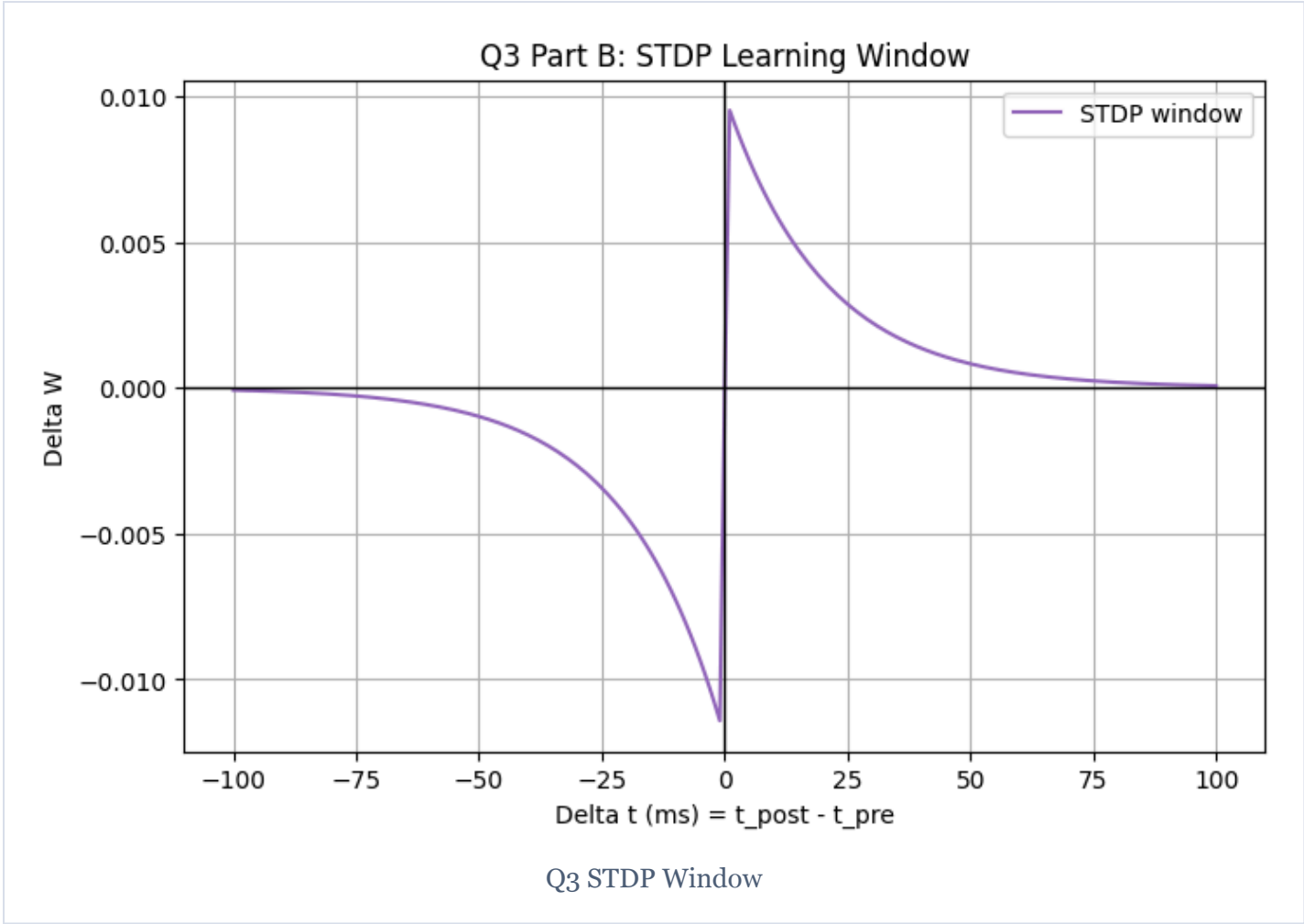
Random case pre spikes: 10 post spikes: 11

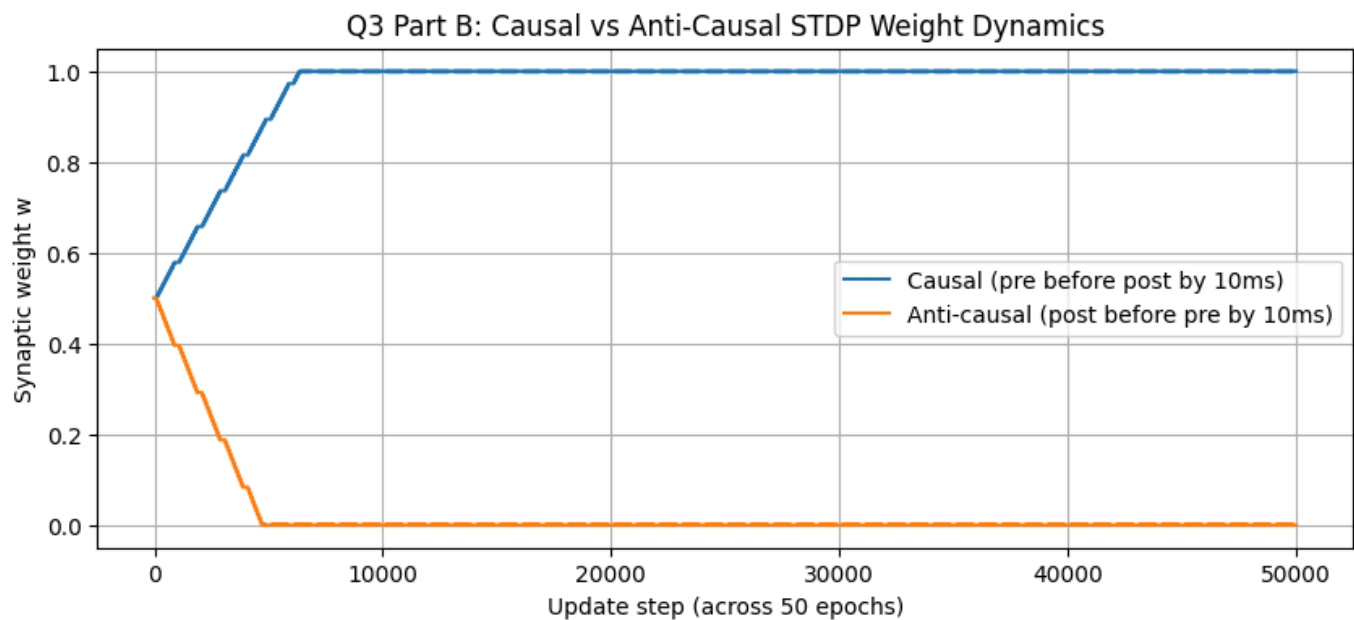
Causal final weight : 1.0

Anti-causal final weight: 0.0

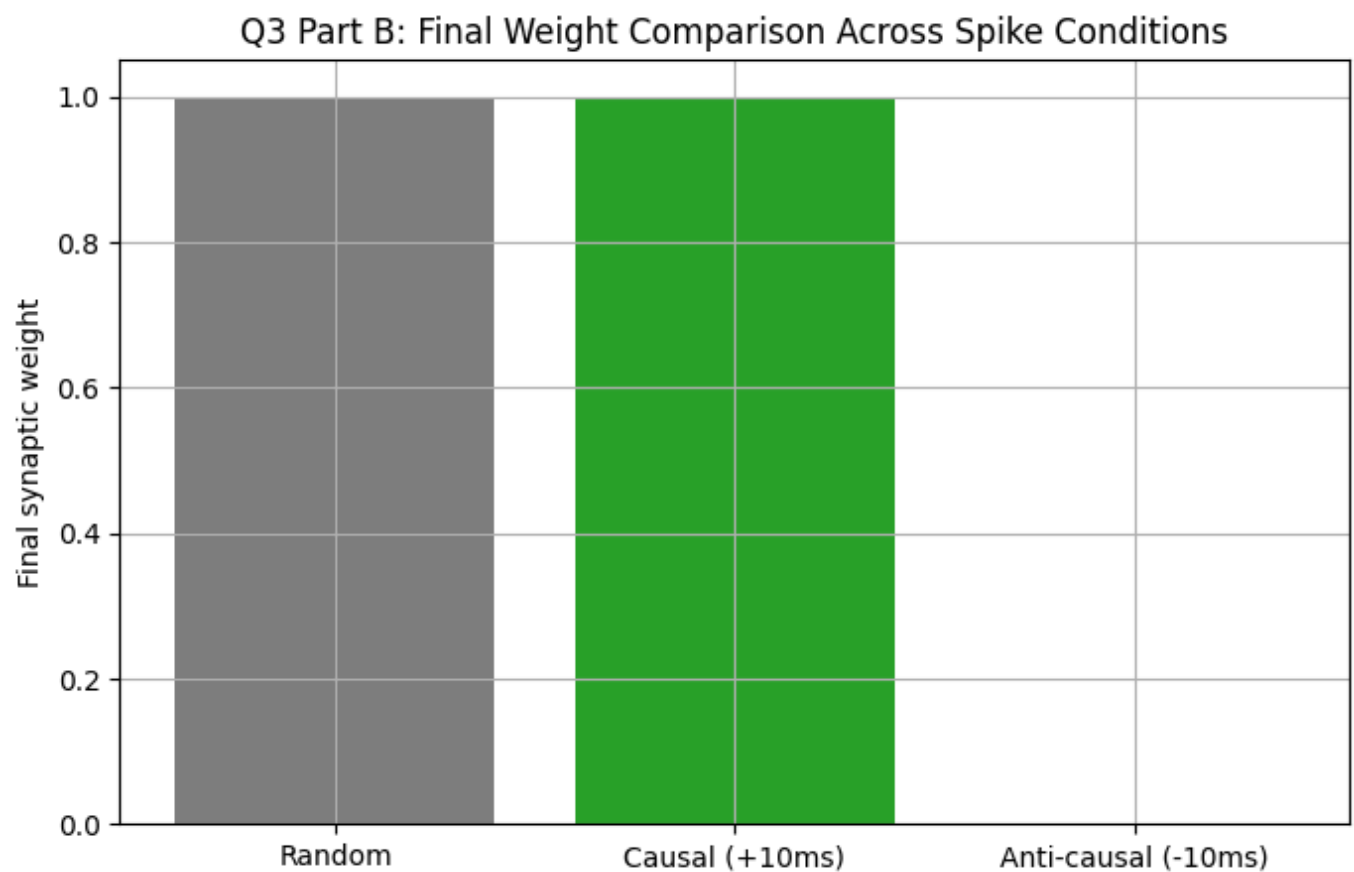
Expected trend check: causal > anti-causal (usually true under standard STDP par

3.2.5 Visual Results





Q3 Causal vs Anti-Causal Trajectories



Q3 Final Weight Comparison

3.2.6 Interpretation

The simulation outcome is consistent with STDP theory:

- causal ordering (pre before post) increases weight,

- anti-causal ordering decreases weight.

This directly demonstrates temporal Hebbian plasticity and its connection to differential Hebbian learning principles.

4. Consolidated Result Summary

Section	Key Quantitative Result	Conclusion
Q1 Task 2.1	Hebbian <code>norm(w)</code> and angle became <code>nan</code>	Unstable / diverging dynamics
Q1 Task 2.2	Oja vs PC1 angle = 0.4779°	Accurate PCA-direction recovery
Q1 Task 2.3	Mean Oja-PCA similarity = 0.8882	Strong alignment with PCA components
Q3 Part B	Causal final $w = 1.0$, Anti-causal final $w = 0.0$	Correct LTP/LTD timing behavior